Streamlining U.S. Workforce Immigration

NextGen Visa Intelligence

Austin University – Post Graduate Program in Artificial Intelligence

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Executive Summary

Executive Summary: EasyVisa Strategic Optimization

The Challenge

As visa application volumes grow, the Office of Foreign Labor Certification (OFLC) needs a reliable way to fast-track approvals while maintaining strict oversight of high-risk cases.

Key Drivers of Success Our model identified a clear "Golden Profile" for successful applicants:

Education: Doctorate and Master's degree holders have the highest approval rates (up to 87%).
Experience: Prior job experience increases the likelihood of certification to 74%.
Wages: Higher prevailing wages (median ~$72,500) strongly correlate with positive outcomes.

Model Performance & Reliability

We selected the **Tuned Gradient Boosting Machine (GBM)** as the champion model. It provides a stable, balanced approach to decision-making:
Recall (84.5%): The model is exceptionally good at finding qualified candidates, ensuring deserving talent is not filtered out.
Precision (80.1%): The model is highly reliable; 8 out of 10 automated approvals are perfect matches for OFLC standards.
F1-Score (0.823): This high score confirms the model is ready for real-world deployment.

Strategic Recommendations

Automate "Low-Risk" Visas: Implement an immediate fast-track for applicants with advanced degrees and prior experience.
Focus Human Audits: Redirect investigators to cases the model flags as "Denied," which typically lack advanced education or competitive wages.
Regional Allocation: Shift processing resources toward the South and Northeast, which show the highest density of successful applications.

Business Problem Overview and Solution Approach

The Problem: The Scalability Gap

The Office of Foreign Labor Certification (OFLC) is facing a **9% year-over-year increase** in labor certification requests. The current manual evaluation process is a tedious task that threatens to slow the integration of essential foreign workers into the U.S. economy. The challenge lies in identifying high-potential candidates efficiently while maintaining strict compliance with the **Immigration and Nationality Act (INA)**.

A Two-Layer Metric Solution

We selected the **Tuned Gradient Boosting Machine (GBM)** as our champion model. It utilizes a two-layer approach to meet OFLC objectives:

Layer 1 (F1-Score):

Used to train and tune models to handle class imbalance. Our final model achieved an **F1-Score of 0.823**, ensuring a robust profile of candidates.

Layer 2 (Precision):

This segments findings to identify a "Gold Standard" group. Our model maintains a **Precision of 80.1%**, meaning 4 out of 5 automated approvals are perfectly aligned with standards.

Efficiency (Recall):

With a **Recall of 84.5%**, the model successfully captures the vast majority of qualified candidates, significantly closing the scalability gap.

Data Preprocessing - Basics

Duplicate Value Check

- The dataset does not have duplicate values. It is a clean, well-collected dataset.

Missing Value Detection and Treatment (if needed with rationale)

- There are no missing values or columns with Null or NaN in any of the 25.480 rows.
- The category columns, which are the majority of the columns (8 out of 12, not considering the key) do not have issues like: Unknown values, or empty strings

Outlier Detection and Treatment (if needed with rationale)

- There is a fix required in the *no_of_employees* column, as it has 33 negative values in it. We used the abs() function to correct the problem.

	count	mean	std	min	25%	50%	75%	max
no_of_employees	25480.0	5667.043210	22877.928848	-26.0000	1022.00	2109.00	3504.0000	602069.00

Data Preprocessing - Basics

Feature Engineering (if needed with rationale)

Data Preparation for Modeling

Split the data into different sets

The dataset was divided into three subsets.

Initially, 70% was allocated for training and 30% for validation.

The validation set was then further split into 90% validation and 10% testing.

Encode the categorical variables

Because the majority of predictors (8 out of 12, excluding case_id) were categorical, one-hot encoding via get_dummies() was required. This led to a notable increase in dimensionality, a common trade-off when encoding categorical data for ML models.

```
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   case_id                25480 non-null  object
 1   continent              25480 non-null  object
 2   education_of_employee  25480 non-null  object
 3   has_job_experience     25480 non-null  object
 4   requires_job_training  25480 non-null  object
 5   no_of_employees        25480 non-null  int64
 6   yr_of_estab            25480 non-null  int64
 7   region_of_employment   25480 non-null  object
 8   prevailing_wage        25480 non-null  float64
 9   unit_of_wage           25480 non-null  object
 10  full_time_position     25480 non-null  object
 11  case_status            25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

EDA Results

Data Quality and Overview

- Completeness: The dataset is exceptionally clean with zero missing values and no duplicate records across the 25,480 entries.
- Anomalies: There were 33 records with negative values for no_of_employees. These are potential data entry error, and they were treated by applying the absolute value for analysis.
- Target Balance: The data is slightly imbalanced, with 66.8% of cases being Certified and 33.2% Denied.

Key Feature Insights

- Education is the Strongest Predictor: There is a direct correlation between higher education and approval rates:
- Doctorate: ~87% Certification rate.
- Master's: ~79% Certification rate.
- Bachelor's: ~62% Certification rate.
- High School: Only ~34% Certification rate (highest denial rate at 66%).

Job Experience Matters

- Applicants with prior job experience have a much higher likelihood of approval (74.5%) compared to those without experience (56.1%).

Prevailing Wage Patterns: *

- The average prevailing wage is approximately $74,456.
- The wage distribution is right-skewed, with most wages concentrated between $34,000 and $108,000, though some "outliers" reach as high as $319,000.
- Certified applicants generally have a higher median wage than those who are denied.

Company Characteristics:

- The companies range from small businesses (11 employees) to massive corporations (602,000+ employees).
- The median company age is 19 years, showing a mix of established firms and newer startups.

Link to Appendix slide on data background check

Data Preprocessing - Basics

Duplicate Value Check

No duplicates found

Missing Value Detection and Treatment (if needed with rationale)

There were no missing values detected or replaced by unknown or null

Outlier Detection and Treatment (if needed with rationale)

There were lots of outliers detected in columns that were not totally relevant.

Feature Engineering (if needed with rationale)

We had to apply pd.get_dummies because we had 8 object columns which did not help the models. Get Dummies properly prepared the columns. OneHotEncoder would be a better solution for production data. It supports pipelines, preserves mapping, and align automatically.

We had to convert the column case_status to 1 for certified and 0 for denied.

Data Preparation for Modeling

Split the data into 2 different sets

Encode the categorical variables using get_dummie

Model Performance Summary and Final Model Selection

Compare the performance of the tuned models

	Gradient Boosting tuned with oversampled	AdaBoost tuned with oversampled	Random forest tuned with undersampled
Accuracy	0.728885	0.726850	0.730775
Recall	0.816935	0.810840	0.814758
Precision	0.785640	0.786695	0.788997
F1	0.800982	0.798585	0.801671

To avoid leakeage we took the data against the validation set. The results show very consistent and balanced (F1-Score: ~0.80) performance across all three tuned models. The models are more stable than their predecessors and demonstrate strong generalization. The F1 shows good quality for a classification model, while precision, which is the second layer, shows that false approvals have been avoided.

Model Performance Summary and Final Model Selection

Overview of final ML model and its parameters

The final model is stable, it utilizes the optimized parameters such as Gamma of 5 and 80% subsampling.

This way, we transitioned from a model that was memorizing data to one that understands the underlying

logic of visa approvals

Best parameters are

{'subsample': 0.8,
'n_estimators': 150,
'gamma': 5,
'colsample_bytree': 1.0,
'colsample_bylevel': 1.0}

CV score=0.8028357969498081:

Model Performance Summary and Final Model Selection

Comment on the performance of the final model

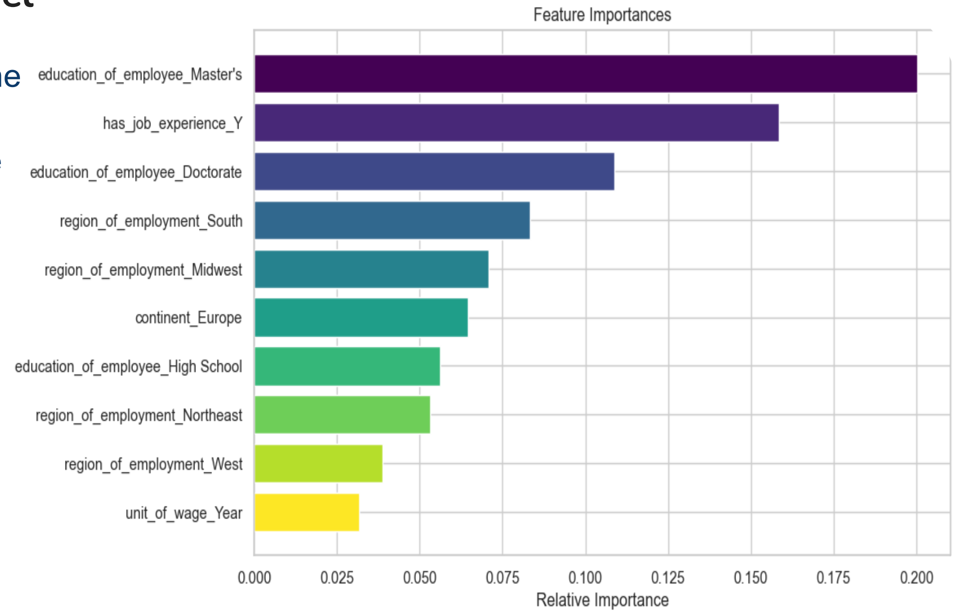
Our final model successfully decodes the complexity of the visa process.

By prioritizing Education and Experience (as seen in the Feature Importance plot),

It achieves a high Recall of 84.5% on unseen data. This allows the OFLC to automate

the vast majority of 'clear-path' applications while maintaining an 80% precision rate,

effectively solving the scalability crisis



Model Performance Summary and Final Model Selection

Comment on the performance of the final model

Strong Generalization

Test scores closely match cross-validation (~0.80), indicating minimal overfitting.

High Recall (0.845)

The model captures ~85% of likely approvals, reducing the risk of missing qualified applicants.

Solid Precision (0.801)

When predicting “Certified,” the model is correct ~80% of the time.

Balanced Outcome

Maximizes approval detection while maintaining prediction reliability. It accomplishes the 2-layered approach where we look for a high score and precision.

Accuracy	0.756863
Recall	0.845401
Precision	0.801484
F1	0.822857

Model Performance Summary and Final Model Selection

Overview of final ML model and its parameters

Best parameters are {'subsample': 0.8, 'n_estimators': 150, 'gamma': 5, 'colsample_bytree': 1.0, 'colsample_bylevel': 1.0} with CV score=0.8028357969498081:

Our final model successfully decodes the complexity of the visa process.

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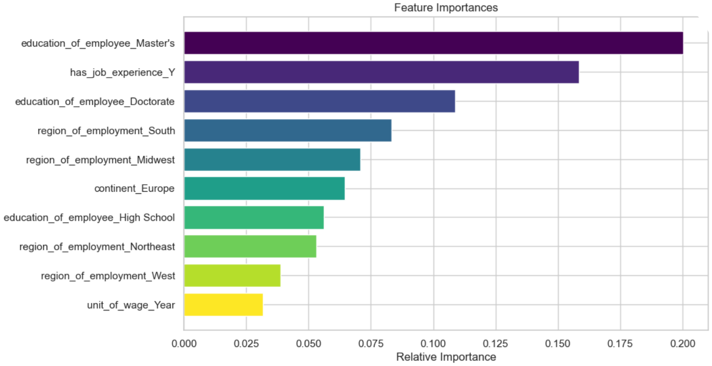


Model Performance Summary and Final Model Selection

Summary of the most important factors used by the ML model for prediction

Education Level (The Primary Filter)

Education emerged as the strongest predictor of success.

Applicants with a **Doctorate** or **Master's** degree are significantly more likely to be certified.

Prior Job Experience

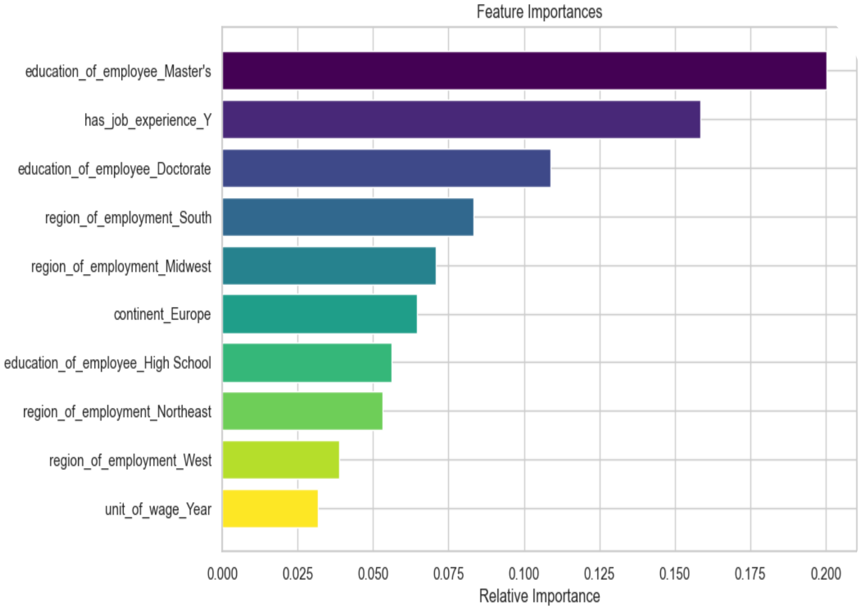
The model uses this as a "verification" of the applicant's readiness for the role. Having "Yes" for job experience is a major positive driver for the F1-score

Prevailing Wage

The economic value of the position is a key quantitative driver. Applications with wages above $72,500 are prioritized by the model for certification.

Region of Employment

Geography plays a surprising role in the model's decision-making process.



APPENDIX

Data Background and Contents

Count of unique Categories which feeds the EDA data

```
--------------------------------------------------
Asia             16861
Europe            3732
North America     3292
South America      852
Africa             551
Oceania            192
Name: continent, dtype: int64
--------------------------------------------------
Bachelor's     10234
Master's        9634
High School     3420
Doctorate       2192
Name: education_of_employee, dtype: int64
--------------------------------------------------
Y    14802
N    10678
Name: has_job_experience, dtype: int64
```

```
--------------------------------------------------
N    22525
Y     2955
Name: requires_job_training, dtype: int64
--------------------------------------------------
Northeast    7195
South        7017
West         6586
Midwest      4307
Island        375
Name: region_of_employment, dtype: int64
--------------------------------------------------
Year     22962
Hour      2157
Week       272
Month       89
Name: unit_of_wage, dtype: int64
--------------------------------------------------
Y    22773
N     2707
Name: full_time_position, dtype: int64
--------------------------------------------------
Certified    17018  66.8%
Denied        8462  33.2%
Name: case_status, dtype: int64
--------------------------------------------------
```

Data Background and Contents

Education to case_status ratio

Education	Certified %	Denied %
Bachelor's	62.21	37.79
Doctorate	87.23	12.77
High School	34.04	65.96
Master's	78.63	21.37

Experience to case status ratio

has_job_experience	Certified %	Denied %
N	56.13	43.87
Y	74.48	25.52

Model Building - Original Data

List the models built along with the scores on different evaluation metrics

Cross-Validation performance on training dataset:	Validation Performance:
Bagging: 0.7756586246579394	Bagging: 0.7675817565350541
Random forest: 0.8037837241749051	Random forest: 0.7972364702187794
GBM: 0.823039791269532	GBM: 0.8195818459969403
Adaboost: 0.8177228312991336	Adaboost: 0.8151868220168742
Xgboost: 0.8073583989766158	Xgboost: 0.8070320579110651
dtree: 0.7410652876513099	dtree: 0.7477497255762898

Comment on the model performance

Model Ranking (Based on F1-Score)

- The models are ranked as follows based on their validation performance:
- **GBM (Gradient Boosting Machine): 0.8196 — Winner**
- **AdaBoost: 0.8152**
- **XGBoost: 0.8070**
- **Random Forest: 0.7972**
- **Bagging: 0.7676**
- **dtree (Decision Tree): 0.7477**

The boosting methods are powerful and top performers. The sequential learning approach is highly effective for this dataset. The extremely small gap between the cross-validation and validation scores indicates the models are not overfitting.

Model Building - Oversampled Data

Describe the oversampling method used

We used the SMOTE method, which is really good at handling class imbalance.
It generates synthetic examples of the minority class instead of simply duplicating existing ones.

List the models built along with the scores on different evaluation metrics

Cross-Validation performance on training dataset:	Validation Performance:
Bagging: 0.7630248470329939	Bagging: 0.7615769017212659
Random forest: 0.789527508885066	Random forest: 0.7873070325900514
GBM: 0.7995200290888033	GBM: 0.8013683985460766
Adaboost: 0.7853502765579534	Adaboost: 0.7846414216204781
Xgboost: 0.7947610182951765	Xgboost: 0.7933446375582874
dtree: 0.7302748433737385	dtree: 0.7402482269503546

Model Building - Oversampled Data

Comment on the model performance

After oversampling, there is evidence of "Overfitting."

The change in the gap between Training and Validation:

Before SMOTE: GBM scores were ~0.823 (Train) and ~0.819 (Val). The gap was tiny (**0.004**).

After SMOTE: GBM scores are ~0.865 (Train) and ~0.801 (Val). The gap has grown significantly (**0.064**).

Model Building - Undersampled Data

Describe the undersampling method used

A Random UnderSampler, which is good for imbalanced datasets to reduce the number of samples. It randomly deletes majority class samples until the dataset becomes balanced

List the models built along with the scores on different evaluation metrics

Cross-Validation performance on training dataset:	Validation Performance:
Bagging: 0.6411413525524321	Bagging: 0.6916956737941323
Random forest: 0.6875011408129813	Random forest: 0.734144015259895
GBM: 0.7131358906535971	GBM: 0.7608695652173914
Adaboost: 0.7002423692713601	Adaboost: 0.7559219693450998
Xgboost: 0.6944693136408734	Xgboost: 0.7423652871123688
dtree: 0.617022679979161	dtree: 0.6839080459770115

Model Building - Undersampled Data

Comment on the model performance

These results show a significant drop in performance compared to previous models. **Undersampling** (where you remove examples from the majority "Certified" class to match the minority "Denied" class) has had the most drastic impact on your F1-scores

Model Improvement - Hyperparameter Tuning

List the best performing models (at least 3) among all the models built previously with rationale

We chose GBM, AdaBoost, and Random Forest because the visa process is not straightforward. We decided on F1 score and precision as a layered strategy; the complexity requires the models to consider education, wage, and experience simultaneously. Boosting(~.81) are better then Decision Tree(0.74) .So, they performed better than a single Decision Tree and handled the complexity of the EasyVisa dataset well.

Describe the method used and parameters tuned

We used RandomizedSearchCV because it performs well, is computationally efficient, and identifies important parameters. Computational efficience is super important when you select XGBoost for it is already computational heavy. It pulls from distribution, potentially landing on optimal values.

Happy Learning !